

Explaining the MeanFlow Equations

Speaker bullets for presentation slides 30–34

The five equations on slides 30–34 build a single argument: *define* the object we want → show *why* it is useful → expose the *catch* → make it *computable* → *train* a network on it. The three-panel figure (Diffusion ≈ 35 calls / FlowCast ≈ 10 / MeanFlow 1) stays fixed while the equation underneath advances.

Slide 30 — Average velocity (definition)

$$u(z_r, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau$$

- **The picture on the right is the whole story:** diffusion curves to the forecast in ≈ 35 calls, FlowCast straightens it to ≈ 10 , MeanFlow wants to jump there in **1**.
- $v(z_\tau, \tau)$ is the **instantaneous** velocity FlowCast already learns — the local tangent at each point of the path.
- u is its **time-average over the interval** $[r, t]$: displacement divided by interval length. It is the single constant velocity that carries you from z_r to z_t .
- Crucially, u is a **property of the flow itself**, not of any network — and as $t \rightarrow r$ it collapses back to v (one evaluation = FlowCast).

Slide 31 — Single-step transport (why we want u)

$$z_t = z_r + (t-r)u(z_r, r, t)$$

- This is just the definition rearranged — but read it as a **sampler**: knowing u turns transport into **one algebraic step**, no ODE solver.
- Set $(r, t) = (0, 1)$ and a **single evaluation maps noise** → **forecast** (the green dot, one call).
- This is where MeanFlow differs from FlowCast in kind, not degree: FlowCast *integrates* many small steps; MeanFlow takes **one exact step**.
- **The catch:** u is defined by an intractable path integral (slide 30), so we cannot regress a network onto it directly. The cost has moved from inference into training.

Slide 32 — The MeanFlow identity (boxed = key result)

$$u(z_r, r, t) = v(z_r, r) + (t-r) \frac{d}{dr} u(z_r, r, t)$$

- We **kill the integral by differentiating instead of integrating**. Multiply slide 30 by $(t-r)$, differentiate in the state-time r (product rule + fundamental theorem of calculus).

- Result relates the field we **want** (u) to the field we **can sample** (v), using only quantities at a **single point of a single trajectory** — the integral is gone.
- The term $(t - r) \frac{d}{dr} u$ is the **correction** that lifts the local velocity v to the interval average u .
- At $t = r$ it reduces to $u = v$ — **plain flow matching is the boundary condition**, which is exactly why MeanFlow is a clean generalisation of FlowCast.

Slide 33 — Expanding the total derivative

$$\frac{d}{dr} u(z_r, r, t) = v(z_r, r) \partial_z u + \partial_r u$$

- The $\frac{d}{dr}$ in the identity is a **total** derivative — the state z_r itself moves along the trajectory as r changes ($dz_r/dr = v$).
- Chain rule splits it into two **partials we can compute**: motion through space ($v \partial_z u$) plus the explicit dependence on the state-time ($\partial_r u$). (t is held fixed, so no ∂_t term.)
- This is the form that maps directly onto a **Jacobian–vector product (JVP)**: one extra forward-mode pass with tangent $(v, 1, 0)$ for (z, r, t) — no higher-order gradients.
- In short: this line is what makes the identity **actually trainable** on a network.

Slide 34 — Training objective

$$\mathcal{L}_{\text{MF}} = \mathbb{E} \left[\text{sg}(w) \left\| u_\theta(z_r, r, t, c_t) - \text{sg}(u_{\text{tgt}}) \right\|_{2,\beta}^2 \right] + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}},$$

$$u_{\text{tgt}} = v + (t - r)(v \partial_z u_\theta + \partial_r u_\theta)$$

- We **regress the network onto the identity’s right-hand side**: the target u_{tgt} is just slide 32 with slide 33 substituted in, and the network **bootstraps its own derivative**.
- c_t **is the conditioning bundle** the network sees alongside the noisy state — $c_t = \text{concat}(X_{t-1}, M_t, I)$: the previous high-res state X_{t-1} , the frozen regression mean M_t (the deterministic forecast this residual is correcting), and the static invariants I (land-sea mask, orography). It is concatenated channel-wise with z_r as network input. The low-res ERA5 field S_t is *deliberately omitted* from the direct path — its synoptic signal already enters through $M_t = F_{\text{reg}}(X_{t-1}, S_t, I)$. The same c_t is used by EDM, FlowCast, and MeanFlow, so only the generative trajectory differs across heads.
- $v = x_1 - x_0$ is the conditional I-CFM velocity (same substitution FlowCast makes, valid in expectation) — **still teacher-free, no diffusion model queried**.
- **stop-gradient (sg)** on the target and on the weight: we do not backprop through the JVP, so training stays first-order and stable.
- $w = (\|\Delta\|_{2,\beta}^2 + \varepsilon)^{-p}$ is adaptive loss weighting (down-weights hard samples); $\|\cdot\|_{2,\beta}$ carries the same per-channel weights $\beta = (1, 1, 1, 2)$ and $\mathcal{L}_{\text{spec}}$ the same **qpepre** spectral term as FlowCast.
- A network at zero loss **satisfies the identity $\rightarrow$ equals the true average velocity**. And on the 75% of the batch where $t = r$, $u_{\text{tgt}} = v$ and this is **literally the FlowCast loss** — so MeanFlow vs FlowCast is an A/B on the objective alone.

How stop-gradient and the JVP fit together (the trick that makes it cheap)

The target $u_{\text{tgt}} = v + (t - r)(v \partial_z u_\theta + \partial_r u_\theta)$ contains a *derivative of the network itself*. Two tools handle it:

- **The JVP computes that derivative — forwards, in one pass.** The term $v \partial_z u_\theta + \partial_r u_\theta$ is exactly the total derivative $\frac{d}{dr} u_\theta$ of slide 33: the network’s Jacobian contracted against the trajectory tangent $(v, 1, 0)$ for the inputs (z, r, t) . A **forward-mode Jacobian–vector product (JVP)** returns precisely this contraction *without ever forming the full Jacobian* — cost $\approx$ one extra forward pass (`torch.func.jvp`).
- **Stop-gradient is what keeps the JVP first-order.** The JVP is itself a (differentiable) function of the weights θ . If the target were left live, backprop through the loss would have to differentiate the JVP w.r.t. θ — a **gradient of a derivative**, i.e. second-order / Hessian-vector terms, expensive and unstable on a 78.7 M-parameter network.
- **So sg freezes the target into a constant.** Wrapping u_{tgt} in stop-gradient (`jvp` run under `no_grad`) turns the right-hand side into a fixed number. The loss becomes an **ordinary first-order regression** $\|u_\theta - \text{sg}(u_{\text{tgt}})\|^2$: gradients flow only through the *prediction* u_θ on the left, never through the derivative target on the right.
- **Together = a bootstrapped regression.** Compute the derivative target cheaply with forward-mode, freeze it, then take a plain MSE step — the network chases a moving target built from a frozen copy of *its own* derivative. (Same idea as the target network in TD-learning, or the stop-grad in consistency models.)
- **Why it matters:** this pairing is what replaces the intractable path integral (slide 30) with **one forward-mode JVP at first-order cost**, and is why MeanFlow trains with the *same loss machinery* as FlowCast. The JVP overhead is confined to the 25% ($r < t$) subset of the batch; the prediction pass runs once over the full batch.

What is $\mathcal{L}_{\text{spec}}$? (the spectral regulariser)

$$\mathcal{L}_{\text{spec}} = \frac{1}{|\mathcal{K}|} \sum_{\kappa \in \mathcal{K}} |\log P_{\text{pred}}(\kappa) - \log P_{\text{target}}(\kappa)|$$

- **What it measures:** the L1 distance between the **radially-averaged log power spectra** of the predicted and target **qppepre** fields. $P(\kappa)$ is the 2D power spectral density averaged over all directions at wavenumber κ (i.e. “how much energy lives at spatial scale $1/\kappa$ ”), and $\mathcal{K}$ is the set of non-zero radial wavenumbers on the grid.
- **Why we need it:** pixel-wise RMSE-type losses reward **smooth, low-amplitude** precipitation — the model minimises error by predicting a “slightly-wet-everywhere” blur and throws away the sharp convective structure we actually care about. The spectral term penalises a **wrong spatial spectrum**, forcing the right amount of fine-scale (high- κ) energy back in.
- **Log of the PSD** because precipitation energy spans orders of magnitude across scales; the log makes the regulariser care about *all* wavenumbers, not just the few most energetic ones.
- **Applied only to qppepre**, on the flow’s *implied clean residual* $\hat{x}_1 = z_r + (1 - r) u_\theta$ (not the raw velocity), and — for MeanFlow — only on the collapsed $t = r$ subset, so it stays directly comparable to the FlowCast term. Weighted by $\lambda_{\text{spec}} = 0.1$.

Presentation note. The same three-panel figure repeats on all five slides while the equation underneath advances. If that is intentional (equations animating in over a fixed visual), these bullets match it. If you prefer, let the coloured dots / trajectory in the figure light up in step with the equation — e.g. highlight the single green MeanFlow dot on slide 31 when you introduce one-step transport.

Suggested content: Experiments / Discussion / Conclusion slides

These are suggested slide skeletons grounded in the thesis numbers (Chapters 4–5). Each “slide” is one idea; pick the subset that fits your time budget. Keep one headline number per bullet — the audience remembers the $\approx 10\times$ speedup, not the third decimal of CRPS.

EXPERIMENTS

E1 — Setup / “How we compare fairly”

- **One question:** can a teacher-free 1–2-call model match the 35-call diffusion baseline? Three heads, **everything else held fixed** — same frozen regression mean, same SongUNet backbone, same σ_{data} , same cleaned log1p dataset.
- **Matched budget = training *samples* seen ($\approx 2\text{ M}$), not steps** (batch sizes differ: EDM 64, FlowCast 96, MeanFlow 112). So any gap reflects the *generative trajectory*, not training compute.
- **Evaluation:** 2022 full validation year (8,759 hours), 5-member ensembles, 10 initial times, +1 h... + 6 h rollouts. All scores denormalised to physical units (mm h^{-1} , K, m s^{-1}).
- **Caveat up front:** single-snapshot validation noise is 10–30% — only call gaps that clear that floor.

E2 — Headline scoreboard (the central result)

- **FlowCast ($K=10$) ties or beats diffusion at $3.7\times$ less cost:** CRPS 0.552 vs 0.554 (a tie), and clearly ahead on winds (-5% u10, -12% v10) and precipitation RMSE (-7%).
- **MeanFlow ($K=2$) posts the best precip hit rates of any head** — CSI-M 0.269 and HSS-M 0.315, both *above* diffusion’s 0.262/0.293 — at $\approx 10\times$ less cost (1.16s vs 11.2s per sequence).
- **The trade is coherent, not a wash:** FlowCast is the *conservative* forecaster (lowest false alarms), diffusion and MeanFlow- $K2$ are more *aggressive* (more hits, more false alarms). Which you want is an operational choice.
- **MeanFlow’s one concession:** rollout CRPS lags by $\approx 10\%$ — a spread (calibration) gap, diagnosed later, not a skill gap.
- Suggested visual: the 4-row scoreboard table or `scoreboard_3way.png`.

E3 — Inference cost (the “money” slide)

- **The cost that motivated the thesis:** a 50-member, 24 h ensemble cycle. Diffusion does $50\times 24\times 35 = 42,000$ network calls per cycle; MeanFlow- $K2$ does just **2,400** (1,200 at $K=1$).
- **Wall-clock (A6000):** EDM $\approx 31.6\text{ min}$ $\rightarrow$ FlowCast- $K10 \approx 10.8\text{ min}$ ($2.9\times$) $\rightarrow$ **MeanFlow- $K2 \approx 4.0\text{ min}$** ($7.9\times$), MeanFlow- $K1 \approx 3.2\text{ min}$ ($9.9\times$).
- **Hardware-independent, and the gap *widens* on faster silicon:** on the H100 the speedups grow to $3.1\times / 8.5\times / 10.8\times$ (diffusion stays compute-bound on its many calls).

- **No second training stage:** these speedups are *intrinsic* to the straight-line / average-velocity model — no distillation, no teacher. “A path worth turning twice: first straightened, then jumped.”
- Suggested visual: `timing_speedup.png` (log-axis bars, both GPUs).

E4 — Where the differences live (precip skill + rollout)

- **Light/moderate rain:** flow heads have a clear false-alarm advantage (FlowCast FAR 0.20 vs diffusion 0.29 at 0.1 mm h^{-1}) — they don’t paint speculative drizzle on dry pixels.
- **Heavy rain ($\geq 5 \text{ mm h}^{-1}$) is hard for everyone** at this budget — a genuine, honest weakness; neighbourhood pooling (recovering displacement) and longer training are the levers.
- **Rollout is stable:** all heads grow error monotonically, none blows up; the cleaned flow heads avoid the legacy baseline’s unphysical `qpepre` excursion at +4 h.
- **Qualitative case (2022-10-15, $\approx 80 \text{ mm h}^{-1}$ peak):** diffusion spreads broad rain, FlowCast damps the intense cores, **MeanFlow-*K*2 keeps the most high-intensity structure at 2 calls** — and is *not* visibly blurrier than FlowCast’s 10-step fields.

E5 — Ablations (optional / backup)

- **log1p encoding is the single biggest preprocessing win:** compresses `qpepre`’s std from 9.12 to ≈ 0.37 and improves *every* channel ($\sim 20\text{--}25\%$ on t2m) — the badly-scaled raw channel otherwise contaminates the shared U-Net trunk.
- **Precip weight w_{pr} :** no clean monotone trend; $w_{\text{pr}}=2.0$ is the all-round default (best t2m, competitive `qpepre`), 1.4 is the wind sweet spot. MeanFlow reuses 2.0 unchanged so the A/B stays clean.
- **Legacy $\rightarrow$ cleaned pipeline** is a large step on its own (CRPS $0.63 \rightarrow 0.55$, CSI-M $0.14 \rightarrow 0.26$), *independent* of the head — MeanFlow adds a $17\times$ end-to-end speedup on top.

DISCUSSION

D1 — What the evidence says

- **Answer to the thesis question: a qualified “yes.”** At 1–2 calls MeanFlow reaches the diffusion baseline’s deterministic *and* categorical skill class, with the best precip hit rates and near-best temperature, at $\approx 1/10$ the cost.
- **FlowCast marks the intermediate point on the same axis:** the straight-line path already recovers most of the diffusion skill ($K=10$, $3.7\times$); the average-velocity jump removes the *last* of the integration cost.
- **The one honest concession is calibration, not skill:** MeanFlow’s rollout CRPS gap is a spread problem (next slide), most of it closed by the second segment.

D2 — Failure modes (be honest)

- **Heavy-rain under-detection** — all heads converge to low CSI above 5 mm h^{-1} at 2 M samples; flow heads’ conservatism costs hits.

- **MeanFlow under-dispersion at $K=1$** — one average-velocity jump lands each noise draw near the conditional mean, narrowing ensemble spread; it *compounds* under autoregression. $\Rightarrow K=2$ is the recommended operating point.
- **FlowCast t2m rollout drift** (1.73 K vs 1.60 K at +6 h) — but MeanFlow does *not* inherit it despite sharing the backbone, so it’s a property of the learned *trajectory*, not the architecture.
- **Three caveats bounding all claims:** single 2 M snapshot (10–30% noise); one regression mean / one domain (Taiwan); the log1p encoding must match between regression and residual head.

CONCLUSION

C1 — Contributions

- **First application of the MeanFlow objective to a weather forecaster:** a single-run, teacher-free average-velocity residual head for StormCast, sampling in 1–2 calls with *no* ODE solver.
- **A controlled three-head comparison** (EDM / FlowCast / MeanFlow) sharing regression mean, backbone, and σ_{data} — so MeanFlow-vs-FlowCast is an A/B on the training objective alone.
- **A precipitation-aware loss & evaluation suite** (channel-weighted L2 + radial log-PSD; RMSE/CRPS/CSI/FSS/HSS/spectral/ rollout) plus log1p and w_{pr} ablations.
- **The headline finding:** diffusion-class skill at a tenth of the cost, teacher-free.

C2 — Future work

- **Train to convergence and re-score** (FlowCast already at ≈ 13.4 M) — the single most important step to harden the provisional headline against snapshot noise.
- **Restore low-NFE rollout spread** — stochastic single-segment sampler / noise re-injection / spread-aware loss, to make the true 1–2-NFE regime probabilistically competitive end to end.
- **Close the heavy-rain gap** (tail-quantile / focal loss on qpepre) and **exploit cheap sampling for large-ensemble calibration** (50–100 members, rank histograms).
- **Isolate the loss recipe** (re-train EDM with the same rain-aware terms) and **test transfer** to another domain (HRRR/CONUS).

C3 — Closing line (one slide, big text)

- Generative modelling is the right inductive bias for convective-scale forecasting; the *multi-step sampler* was the obstacle.
- **The path from noise to weather is a dial worth turning twice: first straightened (FlowCast), then jumped (MeanFlow).**
- Sharp, probabilistic, regional forecasting *fast enough to run between radar volumes* is now within reach — with no teacher.